

Production antipodal last-insertion zero-cross boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope

R-149 proved that an affine Gaussian current owner is controlled by the joint field-current cross synthesis $K = S_x S_v^*$ and required one actual finite production cylinder. This note supplies the first such cylinder. It uses the canonical A1 full-production covariance at the two antipodal Fourier pairs $\{\pm k\}$ and $\{\pm 2k\}$, where

$$k = (\pi/8, 0, 0)$$

on the registered $L = 16$ torus.

The exact advance is deliberately local in the owner chronology. When the final pair $\{\pm 2k\}$ is revealed simultaneously, its coincident field- derivative cross synthesis vanishes. R-149 then makes the conditional expectation of the absolute, full-output, full-reveal endpoint atom nonnegative. This absolute atom spends neither source nor sextic budget.

Three boundaries are load-bearing.

1. The two-point and Fourier-coefficient cross syntheses are nonzero. A nonlocal output projection therefore restores the signed R-149 terms.
2. The relative action contains the difference between final and previous endpoint atoms. Positivity of the absolute final atom does not sign that secant.
3. The earlier $\{\pm k\}$ root sees the later feedback $h_2(\xi_1)$ in its future. R-136 forbids freezing this connection. The complete two-root owner, forest, balanced response, returned low, and global source/sextic windows remain open.

No spatial phase ansatz enters the result. It selects neither BCC nor any alternative phase and has no PDE-replacement implication.

2. The full-production covariance

The A1 quadratic Fourier symbol is

$$A(p) = a(p)I_3 + M, \quad a(p) = r + Z|p|^2 + Y|p|^4, \quad (2.1)$$

where the exact family-lock matrix reconstructed from the registered inputs is

$$M = \begin{pmatrix} 1/10 & -1/20 & -1/20 \\ -1/20 & 13/100 & -1/20 \\ -1/20 & -1/20 & 17/100 \end{pmatrix}. \quad (2.2)$$

Its leading principal minors are positive. For a directed complex mode the A6 convention is

$$\mathbb{E}[\widehat{\Psi}(p)\widehat{\Psi}(p)^\dagger] = 2C(p), \quad C(p) = A(p)^{-1}. \quad (2.3)$$

A7 half-realification therefore gives the six-real covariance

$$\Gamma(p) = \text{diag}(C(p), C(p)), \quad (2.4)$$

with no additional factor one half. Let $G_p G_p^* = \Gamma(p)$.

The determinant of $a\mathbf{I}_3 + M$ is

$$\frac{25000a^3 + 10000a^2 + 1115a + 24}{25000} > 0. \quad (2.5)$$

The executable certificate substitutes the exact decimal A1 inputs into $a(k)$ and $a(2k)$ and verifies positivity before inverting either symbol. This is the full production covariance. The earlier exploratory scalar slice with zero family and lock masses is retained only as a rejected checksum and is not evidence for this theorem.

3. Exact antipodal field and current syntheses

Let

$$J = \begin{pmatrix} 0 & -\mathbf{I}_3 \\ \mathbf{I}_3 & 0 \end{pmatrix}, \quad U(\theta) = \cos \theta \mathbf{I}_6 + \sin \theta J. \quad (3.1)$$

Because $\Gamma(p)$ has the duplicated form (2.4),

$$[J, \Gamma(p)] = 0, \quad U(\theta)\Gamma(p)U(\theta)^* = \Gamma(p). \quad (3.2)$$

Reveal the two signs of one antipodal pair as a single Gaussian block $g = (g_+, g_-) \sim N(0, \mathbf{I}_{12})$. With $\theta_x = p \cdot x$, define

$$S_x(x) = \frac{1}{\sqrt{V}} \begin{bmatrix} U(\theta_x)G_p & U(-\theta_x)G_p \end{bmatrix}, \quad (3.3)$$

$$S_{v,i}(x) = \frac{p_i}{\sqrt{V}} \begin{bmatrix} JU(\theta_x)G_p & -JU(-\theta_x)G_p \end{bmatrix}. \quad (3.4)$$

These are the A1 Fourier field and the physical derivative $\partial_i \leftrightarrow ip_i$ in six-real coordinates.

Theorem 3.1 (coincident cancellation and nonlocal kernel)

For every x ,

$$S_x(x)S_x(x)^* = \frac{2}{V}\Gamma(p), \quad S_{v,i}(x)S_{v,i}(x)^* = \frac{2p_i^2}{V}\Gamma(p), \quad (3.5)$$

and

$$\boxed{K_i(x, x) = S_x(x)S_{v,i}(x)^* = 0.} \quad (3.6)$$

At two distinct points,

$$\boxed{K_i(x, y) = \frac{2p_i}{V}\Gamma(p) \sin\{p \cdot (x - y)\}.} \quad (3.7)$$

Consequently $K_i(y, x) = -K_i(x, y)$ and the kernel is not the zero operator. For the final pair $p = q = 2k$, (3.7) is

$$K_i(x, y) = \frac{4k_i}{V}\Gamma(2k) \sin\{2k \cdot (x - y)\}. \quad (3.8)$$

Proof. Equation (3.2) gives each marginal immediately. In the cross term the $+p$ contribution is $-p_i\Gamma(p)J/V$ and the $-p$ contribution is $+p_i\Gamma(p)J/V$, proving (3.6). At different points the same two terms combine as

$$\frac{p_i}{V}\{U(\theta_y - \theta_x) - U(\theta_x - \theta_y)\}\Gamma(p)J,$$

which reduces to (3.7) because $J^2 = -I_6$. □

In the cosine/sine coefficient basis the same calculation reads

$$S_x^{\text{coef}} = \begin{pmatrix} G_p & 0 \\ 0 & G_p \end{pmatrix}, \quad S_{v,i}^{\text{coef}} = p_i \begin{pmatrix} 0 & G_p \\ -G_p & 0 \end{pmatrix}, \quad (3.9)$$

so

$$K_i^{\text{coef}} = p_i \begin{pmatrix} 0 & -\Gamma(p) \\ \Gamma(p) & 0 \end{pmatrix} \neq 0. \quad (3.10)$$

Thus “ $K = 0$ ” in (3.6) is a coincident local statement, not a projected or coefficient-space statement.

4. The absolute final-endpoint atom

Condition on the strict past before the simultaneous final antipodal pair. All earlier roots and the legal feedback $h_2(\xi_1)$ are then fixed. Write the pointwise field and current as

$$W = w + S_x g, \quad V = v + S_v g, \quad (4.1)$$

where w, v are strict-past measurable. For the complete six-row coefficient map put

$$B(W) = C_6(W)^* C_6(W) \succeq 0, \quad (4.2)$$

and use exactly the once-owned fresh derivative covariance $\Gamma_v = S_v S_v^*$ in the primitive trace. R-149 defines

$$2\mathcal{R}_{\text{raw}} = \|C_6(W)V\|^2 - \text{Tr}\{C_6(W)\Gamma_v C_6(W)^*\}. \quad (4.3)$$

Theorem 4.1 (nonnegative conditional absolute atom)

Assume the positive floor and the Gaussian integrability required by R-149, the full local output identity, simultaneous antipodal revelation, a common fresh-root-independent heat, and no later Gaussian root. Then

$$\boxed{\mathbb{E}[\mathcal{R}_{\text{raw}} \mid \mathcal{F}_{\text{past}}] = \frac{1}{2} \mathbb{E}[v^* B(W) v \mid \mathcal{F}_{\text{past}}] \geq 0.} \quad (4.4)$$

Proof. R-149 Theorem 3.1 gives

$$2\mathbb{E}\mathcal{R}_{\text{raw}} = \mathbb{E}\{v_m B_{mn} v_n + 2v_m K_{an} \partial_a B_{mn} + K_{am} K_{bn} \partial_{ab} B_{mn}\}. \quad (4.5)$$

The fresh-pair synthesis is precisely (3.3)–(3.4), so (3.6) removes both K terms. Equation (4.2) proves the remaining sign. □

The theorem concerns conditional expectation, not pointwise positivity of the random square-minus-trace atom. R-107’s complete one-pair scalar packet has mean zero but a negative pointwise infimum, consistently with (4.4).

At the final full reveal, $P_f = I$ and $\mathcal{V}_{\text{future}} = 0$. With all visits recombined, the R-125/R-149 coordinate relation therefore gives

$$\mathcal{P}_{\text{comp}}^{\text{abs}} = \mathcal{R}_{\text{raw}}^{\text{abs}}. \quad (4.6)$$

The absolute atom (4.6) may be certified with the exact suballocation

$$\eta_{\text{abs},2} = 0, \quad \zeta_{\text{abs},2} = 0. \quad (4.7)$$

No source or terminal sextic is spent to lower-bound this atom.

5. Why the relative secant is still open

The old-chart action does not contain only (4.6). It contains

$$\Delta \mathcal{P}_{\text{comp}} = \mathcal{P}_{\text{comp}}(Z_*) - \mathcal{P}_{\text{comp}}(Z_{\text{previous}}). \quad (5.1)$$

Positivity of the two absolute endpoints does not order them. With constant $B = 1$, previous predictable current $v_0 = 2$, and final predictable current $v_* = 1$, (4.4) gives the two positive atoms 2 and 1/2, but

$$\Delta \mathcal{P}_{\text{comp}} = \frac{1}{2} - 2 = -\frac{3}{2}. \quad (5.2)$$

This fixture is an owner-coordinate scope test, not a production adverse direction. It proves that (4.7) cannot be assigned to the relative secant without computing its baseline cross.

Likewise, a nonlocal output projector introduces double spatial integrals and sees (3.7). The K -derivative terms in (4.5) then return. An exact normalized circle fixture makes this boundary explicit. Let

$$W = a \cos x + b \sin x, \quad V = -a \sin x + b \cos x, \quad a, b \sim N(0, 1), \quad (5.3)$$

and take the scalar row $C(W) = W$. If D_P denotes the unhalved projected square minus its correctly projected primitive trace, direct Fourier calculation gives

$$D_0 = -\frac{a^2 + b^2}{4}, \quad D_2 = \frac{(a^2 + b^2)^2}{8} - \frac{a^2 + b^2}{4}. \quad (5.4)$$

Hence $\mathbb{E}D_0 = -1/2$ and $\mathbb{E}D_2 = +1/2$, while the exhaustive recombination has expectation zero. The full-output identity cannot be replaced by singleton output positivity; this is the same contraction-connected boundary that R-107 found in its one-pair cluster.

6. The legal two-root chart and remaining connection

Let ξ_1 contain the simultaneous $\{\pm k\}$ pair and ξ_2 the simultaneous $\{\pm 2k\}$ pair. The R-119 minimal legal chart is

$$Z_0 = S_1 \xi_1 + S_2 \xi_2, \quad Z_* = Z_0 + S_1 h_1 + S_2 h_2(\xi_1), \quad (6.1)$$

with h_1 strict past and $h_2(\xi_1)$ revealed before root two. For root two, h_2 belongs to w, v in (4.1), so Theorem 4.1 is genuinely adapted and not a frozen-feedback approximation.

The normalized source norm of a linear diagnostic $h_{2,\pm} = tHg_{1,\pm}$ is

$$X(h_2) = 2t^2 \|H\|_{\text{HS}}^2, \quad \frac{9}{20}X(h_2) = \frac{9}{10}t^2 \|H\|_{\text{HS}}^2. \quad (6.2)$$

This checksum fixes the two-sign factor but is not spent in (4.7).

Root one is different. Its future contains root two and the map $h_2(\xi_1)$. Differentiating or conditioning through that future heat produces the R-136 feedback connection. Freezing h_2 changes the owner; the registered two-root fixture loses an exact correction of three in its normalization. R-150 does not recompute this earlier-root block.

There is no abstract positivity extension to that earlier-root geometry. On the normalized circle, reveal $a, b \sim N(0, 1)$ and let a later causal feedback add $h = \lambda a \cos x$. At $x = \pi/4$ and $\lambda = 1$, the terminal field and current synthesis rows have equal variance 5/2 and cross covariance $K = -3/2$. For the positive Gram $B(w) = e^{-w^2}$ and zero past current, R-149 gives the exact unhalved owner

$$K^2 \mathbb{E}B''(W) = -\frac{\sqrt{6}}{8} < 0. \quad (6.3)$$

This is a method no-go for transporting the last-root sign through future feedback, not a counterexample to the complete production action: source, sextic, forest, low, and the other owners have not been added.

For the declared endpoint diagnostic the output projector is the identity, the complement is zero, and the optional covariance split has $\Gamma_{\text{low}} = 0$. Those declarations do not prove that the historical complete-low owner of the full production filtration vanishes. They only remove it from this finite endpoint sublemma. The full R-063 forest incidence, balanced response, and relative source/sextic aggregation must be restored before testing

$$\eta < \frac{5}{11}, \quad \zeta < \frac{13}{50}. \quad (6.4)$$

7. Failed routes and reduced next target

The proof-first audit records four reusable corrections.

1. An exploratory scalar-slice covariance used the N001 scalar value r_0 and omitted both family masses and locking. It was rejected before promotion. Equations (2.1)–(2.5) replace it with the A1 full-production matrix covariance.
2. Coincident cancellation (3.6) cannot be moved through a nonlocal projector. Equations (3.7)–(3.10) preserve the exact missing cross data.
3. The absolute endpoint theorem (4.4) is not the relative secant (5.1). Equation (5.2) prevents a zero-budget overclaim.
4. The last-root sign cannot be transported across an earlier root's future-dependent feedback. The exact fixture (6.3) is negative, so the earlier connection must be computed rather than frozen or inferred.

The next owner calculation stays in the same chart and has exactly two signed targets:

1. compute the final-root relative endpoint secant with its baseline cross;
2. compute the earlier-root common-heat derivative with the full $h_2(\xi_1)$ connection, then restore trace, bracket, future variance, forest, balanced, and historical-low incidence once.

Only their recombination with Theorem 4.1 may be tested against (6.4).

8. Devil's-advocate review

1. **Objection:** $K(x, x) = 0$ **proves every Fourier-output owner is nonnegative. UPHELD against that inference.** The exact two-point kernel (3.7) and coefficient matrix (3.10) are nonzero. Only the local full-output atom is covered.
2. **Objection:** **full reveal alone implies positivity. DISMISSED as the proof offered here.** Full reveal identifies $\mathcal{P}_{\text{comp}} = \mathcal{R}_{\text{raw}}$ but does not sign it. The sign in (4.4) uses the separately proved antipodal $K(x, x) = 0$ and $B \succeq 0$.
3. **Objection:** **the random atom is pointwise nonnegative. UPHELD against that claim.** The theorem is a conditional expectation identity. R-107 supplies a negative pointwise complete-cluster value.
4. **Objection:** **zero budget for the absolute final atom closes its relative insertion. UPHELD against that claim.** Equation (5.2) shows that the difference of positive endpoints can be negative.

5. **Objection: revealing $+2k$ and $-2k$ sequentially gives the same proof. UPHELD against that claim.** Cancellation in (3.6) uses the simultaneous pair, equal antipodal covariance, and a common-even regulator. Each sign separately has nonzero coincident cross synthesis.
6. **Objection: the final-root theorem also closes root one. UPHELD against that claim.** Root one owns future-dependent feedback; R-136 proves that freezing it deletes a load-bearing connection.
7. **Objection: this result favors or excludes a spatial phase. UPHELD against that inference.** No phase ansatz or energy comparison appears anywhere in the theorem.

External review is invited for the A6/A7 factor two, simultaneous antipodal root convention, derivative signs, volume factor, two-point kernel sign, full-output versus future-reveal projectors, absolute-versus-relative owner distinction, and the untouched R-136 feedback connection.

9. Result footer and reproduction

Dependencies: A1, A6, A7, R-107, R-119, R-125, R-136, R-149.

Reproduction command:

```
& E:\Dev\TECT.venv\Scripts\python.exe '
codes/foundations/a13_classii_production_antipodal_last_insertion_zero_cross_boundary_verify.py
```

Result ID: A13-CLASSII-PRODUCTION-ANTIPODAL-LAST-INSERTION-ZERO-CROSS-BOUNDARY; Ledger ID: R-150.

Precise statement: exact A1 antipodal S_x , S_v , coincident $K=0$ and nonlocal

$K(x,y)$; nonnegative conditional absolute final full-output endpoint atom.

Scope: fixed finite root cylinder; simultaneous final antipodal pair; positive floor; full local output and full reveal; fresh-root-independent common heat.

Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.

Expected output: primary and independent PASS; integrated verifier PASS.

Falsification gate: any production covariance, Fourier factor, derivative sign, cross kernel, Stein factor, source norm, scope, hash, PDF, or release check fails.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no relative final secant, earlier-root future-feedback connection, nonlocal projected owner, complete forest/balanced/low aggregate, T-050, A13, Nelson, removals, measure, phase, PDE, or Sector-A closure.

Proof complete: false; T-050 closed: false; Sector A closed: false.

Next required action: compute the final relative secant and earlier-root future-feedback connection in this same production chart, then reassemble all owners once and test $\eta < 5/11$ and $\zeta < 13/50$.